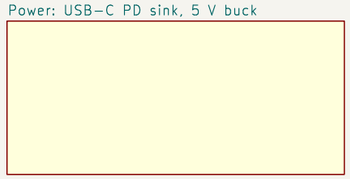
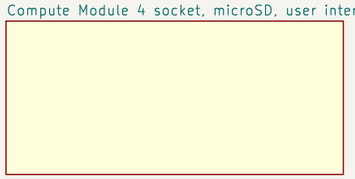
VoxHub CM4 — Home Assistant voice hub
USB-C PD (9–20 V) -> 5.1 V buck -> CM4 Lite + USB hub + audio + Zigbee

Power: USB-C PD sink, 5 V buck



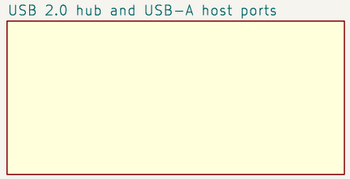
File: power.kicad_sch

Compute Module 4 socket, microSD, user interface



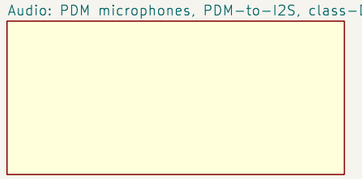
File: cm4.kicad_sch

USB 2.0 hub and USB-A host ports



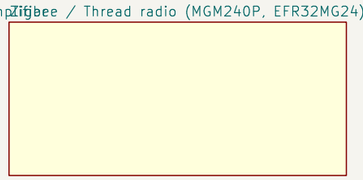
File: usb.kicad_sch

Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier



File: audio.kicad_sch

Zigbee / Thread radio (MGM240P, EFR32MG24)



File: zigbee.kicad_sch

H1
M2.5 board mount

H2
M2.5 board mount

H3
M2.5 board mount

H4
M2.5 board mount

FID1
Fiducial

FID2
Fiducial

FID3
Fiducial

H5
CM4 spacer M2.5x1.5

GND

H6
CM4 spacer M2.5x1.5

GND

H7
CM4 spacer M2.5x1.5

GND

H8
CM4 spacer M2.5x1.5

GND

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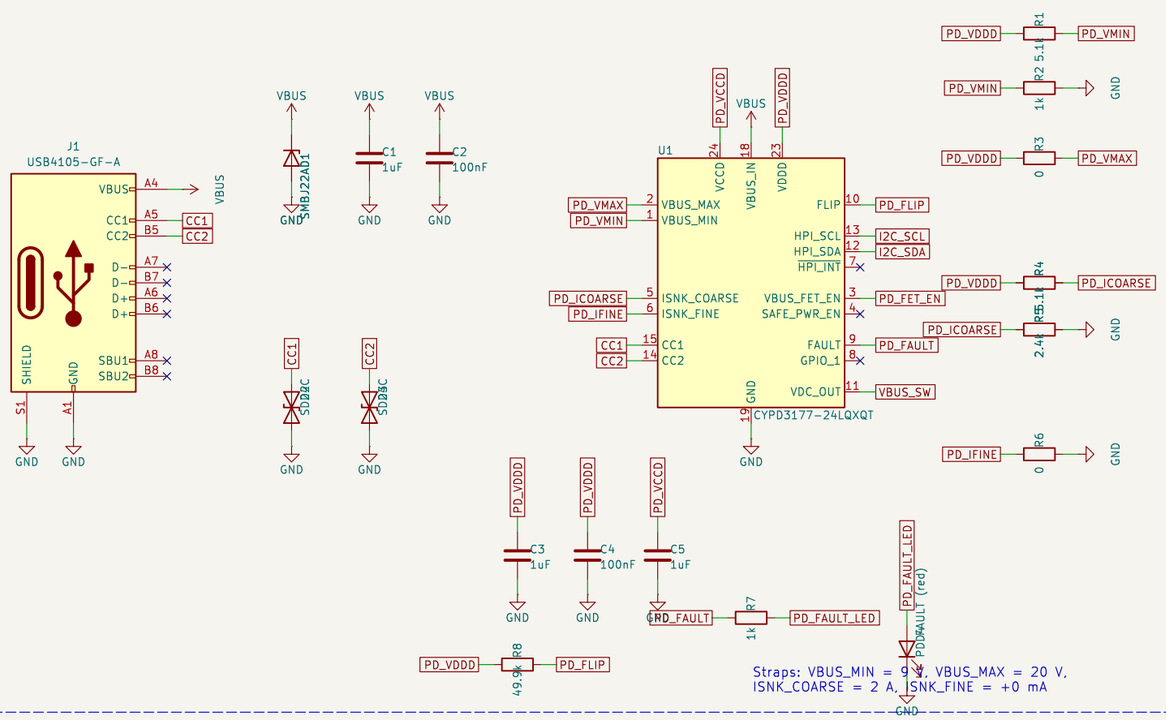
VoxHub (open hardware)

Sheet: /
File: voxhub.kicad_sch

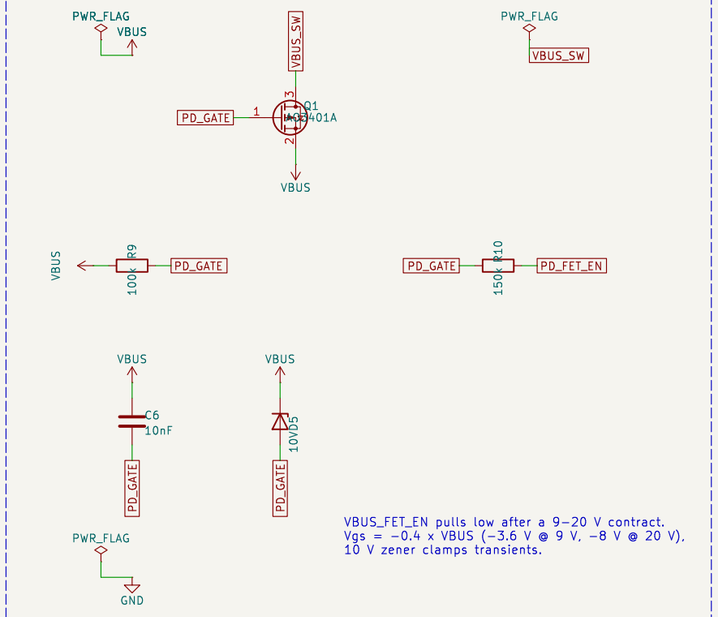
Title: VoxHub CM4 — Home Assistant voice hub carrier

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/6

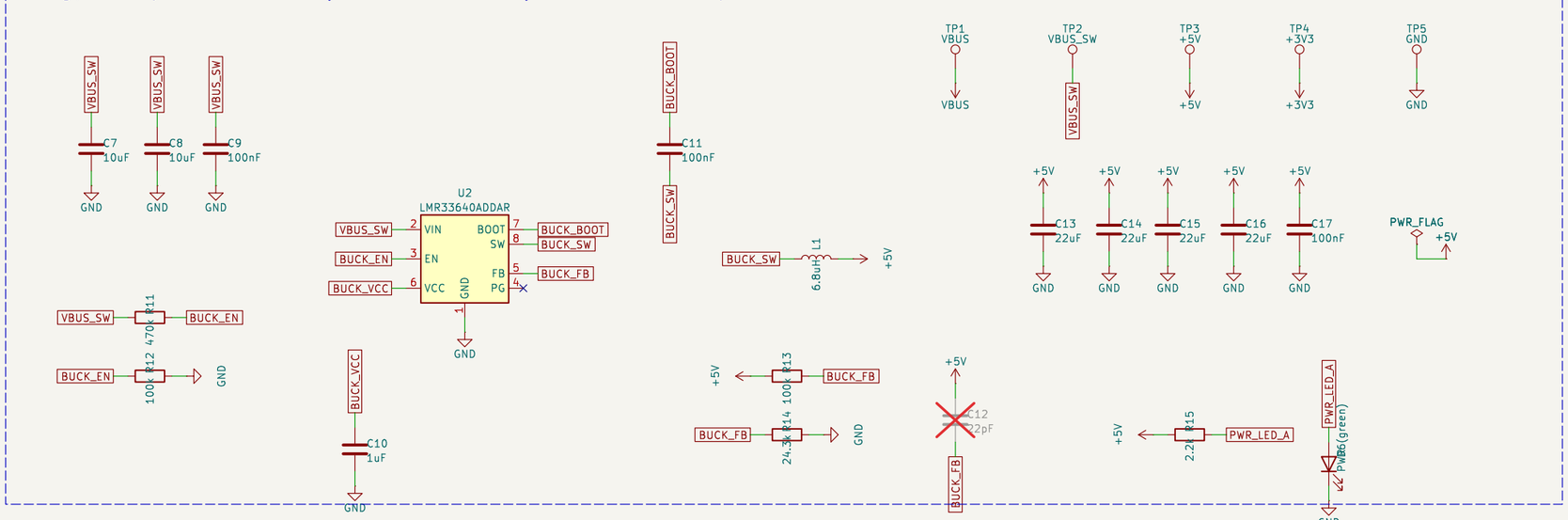
USB-C input and PD sink controller (CYPD3177 'BCR', resistor-strapped: 9–20 V, 2 A minimum)



VBUS load switch: P-FET driven by the BCR's active-low, slew-controlled gate drive



5 V synchronous buck (LMR33640, 400 kHz) — UVLO 7.0 V keeps it off at vSafe5V



Straps: VBUS_MIN = 9 V, VBUS_MAX = 20 V, ISNK_COARSE = 2 A, ISNK_FINE = +0 mA

Generated by hardware/gen — do not edit by hand

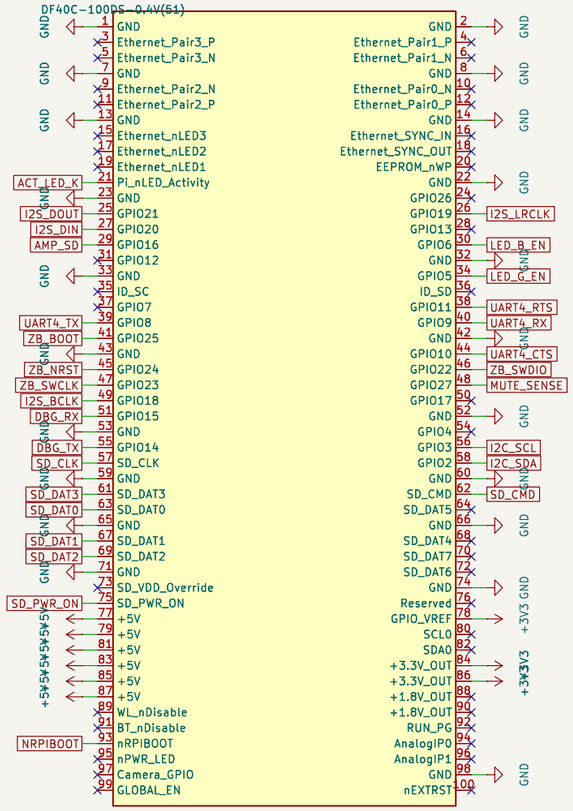
VoxHub (open hardware)

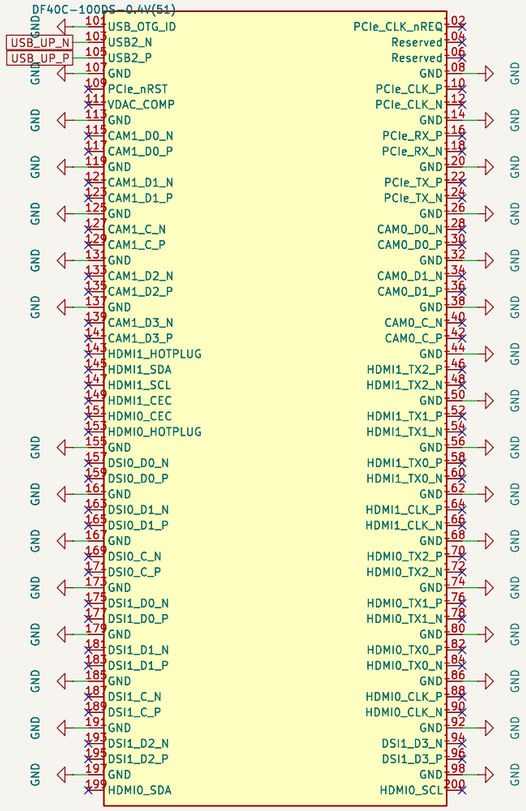
Sheet: /Power: USB-C PD sink, 5 V buck/

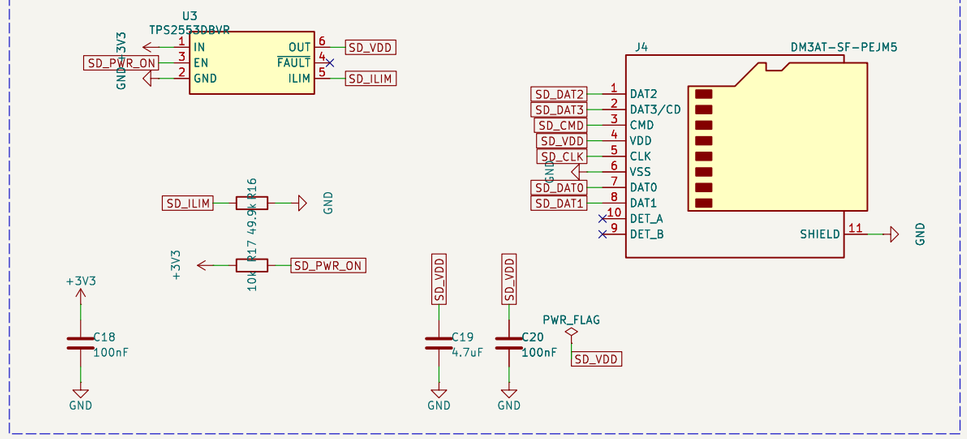
File: power.kicad_sch

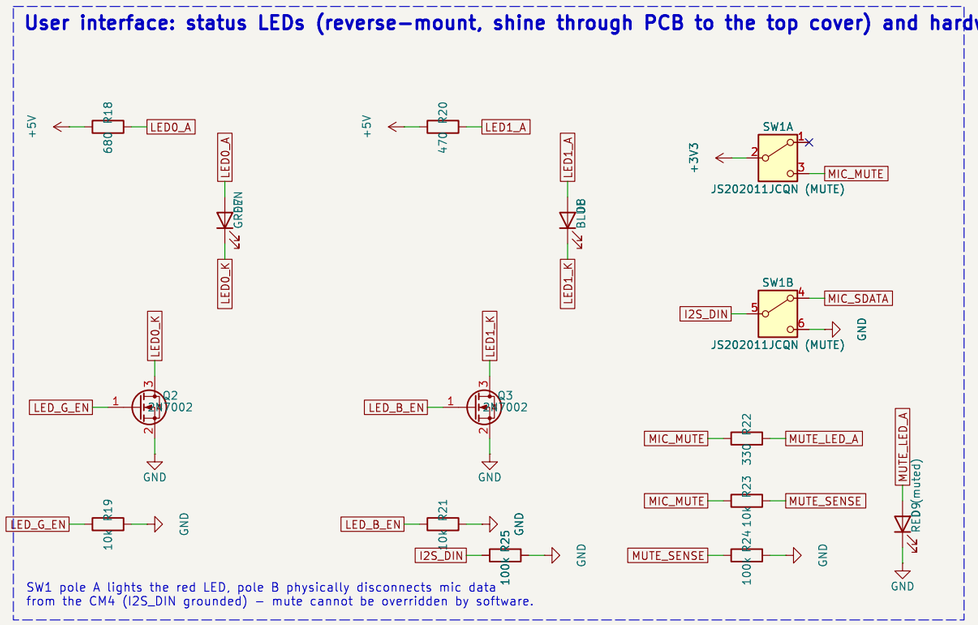
Title: Power: USB-C PD sink, 5 V buck

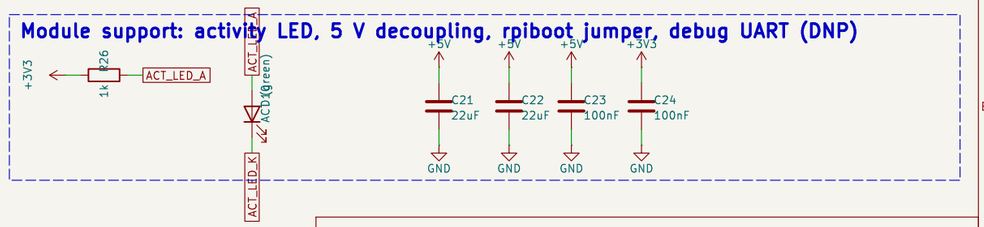
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/6

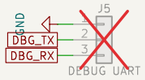














Module: Raspberry Pi CM4 Lite, wireless (CM4104000 / CM4102000 / CM4108000)
GPIO map: see README. USB_OTG_ID tied low = host.

Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)

Sheet: /Compute Module 4 socket, microSD, user interface/
File: cm4.kicad_sch

Title: Compute Module 4 socket, microSD, user interface

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 3/6

USB2514B hub: upstream = CM4 USB2, ports 1–2 = USB-A, ports 3–4 disabled by strap

U4
USB2514B-I/M2

VBUS_DET
USBDM_UP
USBDP_UP
RESET_N
TEST
RBIAS
XTALIN/CLKIN
XTALOUT
PLLFILT
CRFILT
SCL/SMBCLK/CFG_SEL0
SDA/SMBDATA/NON_REM1
SUSP_IND/LOCAL_PWR/NON_REM0
HS_IND/CFG_SEL1
VDDA33
VDD33
VSS
USBDM_DN1/PRT_DIS_M1
USBDP_DN1/PRT_DIS_P1
PRTPWR1/BC_EN1
OCS_N1
USBDM_DN2/PRT_DIS_M2
USBDP_DN2/PRT_DIS_P2
PRTPWR2/BC_EN2
OCS_N2
USBDM_DN3/PRT_DIS_M3
USBDP_DN3/PRT_DIS_P3
PRTPWR3/BC_EN3
OCS_N3
USBDM_DN4/PRT_DIS_M4
USBDP_DN4/PRT_DIS_P4
PRTPWR4/BC_EN4
OCS_N4

HUB_RBIAS R27 12k GND
HUB_PLLFILT C27 100nF GND
HUB_CRFILT C28 100nF GND
HUB_RST_N C29 100nF GND
+3V3 R28 10k HUB_RST_N
HUB_VBUSDET R29 100k +3V3
HUB_CFG0 R30 100k GND
HUB_CFG1 R31 100k GND
HUB_NONREM0 R32 100k GND
HUB_NONREM1 R33 100k GND
HUB_XI Y1 24MHz HUB_XO
HUB_XI C25 27pF GND
HUB_XO C26 27pF GND

+3V3 R34 10k HUB_DIS3M
+3V3 R35 10k HUB_DIS3P
+3V3 R36 10k HUB_DIS4M
+3V3 R37 10k HUB_DIS4P

C30 100nF C31 100nF C32 100nF C33 100nF C34 100nF C35 100nF C36 4.7uF

Downstream ports: per-port 1.2 A current-limited switch, ESD protection, stacked USB-A

U5
TPS2553DBVR
IN EN GND OUT FAULT ILIM
USB_PWR1 +5V GND
USB1_VBUS USB_OC1_N USB1_ILIM

U7
USB4105-2SC6
USB1_N USB1_P USB1_VBUS GND

USB1_ILIM R38 20k GND
+3V3 R39 10k USB_OC1_N

+5V C37 4.7uF GND
USB1_VBUS C38 100uF GND
USB1_VBUS C39 100nF GND
PWR_FLAG USB1_VBUS

J6
61400826021
VBUS1 VBUS2 D1+ D1- D2+ D2- Shield GND1 GND2
USB1_VBUS USB2_VBUS USB1_P USB1_N USB2_P USB2_N GND

U6
TPS2553DBVR
IN EN GND OUT FAULT ILIM
USB_PWR2 +5V GND
USB2_VBUS USB_OC2_N USB2_ILIM

U8
USB4105-2SC6
USB2_N USB2_P USB2_VBUS GND

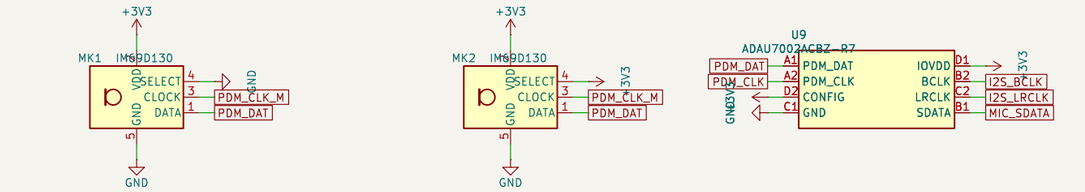
USB2_ILIM R40 20k GND
+3V3 R41 10k USB_OC2_N

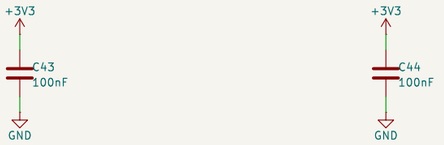
+5V C40 4.7uF GND
USB2_VBUS C41 100uF GND
USB2_VBUS C42 100nF GND
PWR_FLAG USB2_VBUS

Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)
Sheet: /USB 2.0 hub and USB-A host ports/
File: usb.kicad_sch
Title: USB 2.0 hub and USB-A host ports

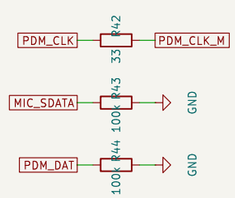
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 4/6

Stereo MEMS microphone array (bottom port through PCB) and PDM -> I2S converter





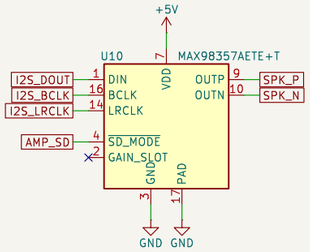
MK1 SELECT=GND (left), MK2 SELECT=VDD (right); 70 mm apart

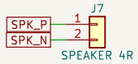


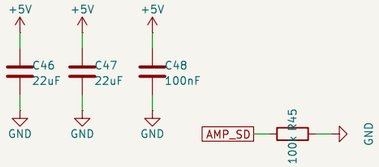


CONFIG = IOVDD -> I2S output; BCLK = 64 fs, PDM clock = 64 fs (3.072 MHz @ 48 kHz)

Speaker amplifier (MAX98357A, I2S in, 3.2 W into 4 ohm from 5 V)







SD_MODE driven by GPIO16 (3.3 V = left channel, low = shutdown). GAIN_SLOT open = 9 dB.

Generated by hardware/gen — do not edit by hand

VoxHub (open hardware)

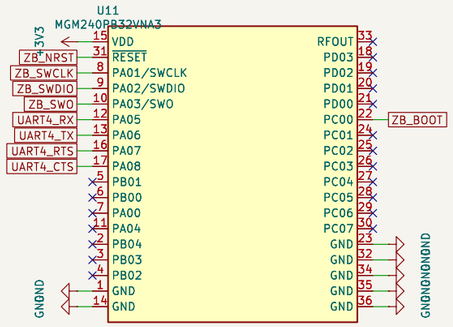
Sheet: /Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier/

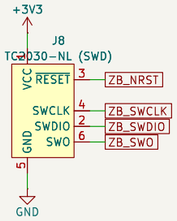
File: audio.kicad_sch

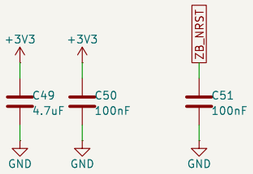
Title: Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier

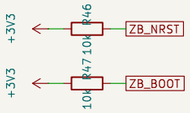
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 5/6

MGM240P network co-processor on CM4 UART4 (GPIO8–11, hardware flow control) + SWD









UART: PA05 = module TX -> GPIO9 (RXD4), PA06 = module RX <- GPIO8 (TXD4).
PA07 = module CTS <- GPIO11 (RTS4), PA08 = module RTS -> GPIO10 (CTS4).
PC00 = bootloader entry, active low (GPIO25), RESET = GPIO24. SWD also wired to GPIO22/23
so the CM4 can flash the radio in-system (openocd bit-bang).

Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)

Sheet: /Zigbee / Thread radio (MGM240P, EFR32MG24)/
File: zigbee.kicad_sch

Title: Zigbee / Thread radio (MGM240P, EFR32MG24)

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 6/6